TAYLOR'S UNIVERSITY
Wisdom · Integrity · Excellence

SCHOOL OF COMPUTER SCIENCE

ITS61504 Data Mining

GROUP ASSIGNMENT-3

HAND OUT DATE AND TIME: July 22, 2026, 11:59 PM

Instructions to students:

- The assignment should be attempted in groups.
- Complete this cover sheet and attach it to your submission – this should be your first page.

Module Learning Outcome:
MLO3 - Work collaboratively to design, execute, and complete a predictive modelling project, demonstrating accountability, autonomy, and ethical responsibility

Student declaration:	
I declare that: ▪ ***I understand what is meant by plagiarism.*** ▪ ***The implication of plagiarism and usage of AI generative tool have been explained to us by our lecturer.*** ***This project is all our work, and I have acknowledged any use of the published or unpublished works of other people.***	
NAME **(change accordingly for group or individual assignment)**	
Name	**Student ID**
1. Group Leader's Name	

Avoid copy and paste job in your report and it is considered as plagiarism. Plagiarism in all forms is forbidden. Students who submit plagiarized assignment will deserve 0 marks.

Assessment Criteria

Assessment Task	Weightage	MLO Assessed	Assessment Instrument	Topics	Week	MQF 2.0
Group Project MLO3	40%	1	N/A	1-12	14	3f

TAYLOR'S UNIVERSITY
Wisdom · Integrity · Excellence

计算机科学学院

ITS61504 数据挖掘

小组作业-3

发布日期和时间：2026年7月22日 晚上11:59

给学生的说明：

- 作业应分组完成。
- 请完成此封面表并将其附在提交材料上——这应为您的第一页。

模块学习成果：
MLO3 - 协作设计、执行并完成一个预测建模项目，展现责任感、自主性和道德责任

学生声明：	
我声明： ▪ 我明白剽窃的含义。 ▪ 剽窃的后果以及使用***AI***生成工具的情况已被我们的讲师解释清楚。这个项目完全是我们的作品，我已确认对任何已发表或未发表的其他人的作品的使用。	
姓名（根据小组或个人作业情况修改）	
Name	**学号**
1. 小组组长姓名	

避免在报告中复制粘贴工作，这将被视为剽窃。所有形式的剽窃都是被禁止的。提交剽窃作业的学生将得到*0*分。

评估标准

评估 Task	权重	MLO 已评估	评估 工具	主题	Week	MQF 2.0
小组项目 MLO3	40%	1	N/A	1-12	14	3f

MLO3: Work collaboratively to design, execute, and complete a predictive modelling project, demonstrating accountability, autonomy, and ethical responsibility

C1 = Knowledge & Understanding, **C2** = Cognitive Skills, **C3A** = Practical Skills, C3B = Interpersonal Skills, **C3C** = Communication Skills, **C3D** = Digital Skills, **C3E =** Numeracy Skills**, C3F** = Leadership, Autonomy & Responsibility, **C4A** = Personal Skills, **C4B** = Entrepreneurial Skills, **C5** = Ethics & Professionalism

Part-I: Assignment Submission Requirements (TOTAL: 40 Marks)

1. Project Report (Softcopy in PDF and DOC)
a. Cover page
b. Marking Rubric
c. A clear work breakdown structure to describe what each member is doing
d. Report Writing Guidelines:

The report should be structured in a coherent and scholarly manner, covering the following key sections:

Question 1: Introduction and Purpose of Study (4 Marks)

a) Provide the general context of the study/project. (1 Marks)
b) Clearly define the problem statement and explain how/why data mining is used. (2 Marks)
c) Describe the business or research motivation and expected outcomes. (1 Marks)

Question 2: Description of Data (4 Marks)
a) Describe the source of the dataset. (1 Marks)
b) Explain the type and structure of the data. (1 Marks)
c) Identify key attributes/features of the dataset. (1 Marks)
d) Present a representative sample (up to 1 page) to illustrate data characteristics.(1 Marks)

Question 3: Data Problems and Pre-Processing (4 Marks)
a) Identify issues in the dataset (e.g., missing, noisy, inconsistent data). (1 Marks)
b) Explain the preprocessing techniques applied (e.g., cleaning, transformation, normalization). (2 Marks)
c) Provide before-and-after comparisons to demonstrate improvements. (1 Marks)

Question 4: Data Mining Techniques (4 Marks)
a) Identify the data mining technique(s) used (e.g., Apriori, classification, clustering). (1 Marks)
b) Justify why the technique(s) were selected for the problem and dataset. (1 Marks)
c) Briefly describe how the technique(s) were implemented. (2 Marks)

Question 5: Model Validation (4 Marks)
a) Describe how the model was evaluated. (1 Marks)
b) Identify the metrics used (e.g., accuracy, support, confidence, lift). (1 Marks)
c) Explain validation techniques or experimental design used to ensure reliability. (2 Marks)

MLO3：协作设计、执行并完成一个预测建模项目，展示责任感、自主性和道德责任

C**1** = 知识 & 理解，C**2** = 认知技能，C3A = 实践技能，C3B = 人际技能，C3C = 沟通技能，C3D = 数字技能，C3E **=** 数理技能，C3F = 领导力、自主性与责任感，C4A = 个人技能，C4B = 创业技能，C**5** = 伦理与专业性

第一部分：作业提交要求（总分：40分）

1. 项目报告（PDF和DOC格式软拷贝）
a. 封面页
b. 评分标准
c. 清晰的工作分解结构，用于描述每个成员的工作内容
d. 报告撰写指南：

报告应采用连贯的学术方式结构化，涵盖以下关键部分：

问题 1：研究介绍和目的（4 分）

a) 提供研究/项目的一般背景。（1 分）
b) Clearl y define the p问题描述和示例 p说明如何/何时 y 数据挖掘 g 被使用。 (2 分)
c) 描述业务或研究动机及预期成果。 (1 分)

问题 2：数据描述 (4 分)
a) 描述数据集的来源。(1分)
b) 解释数据的类型和结构。(1分)
c) Ide..d) 提供一个有代表性的样本（最多1页）,分以说明数据特征。(1分)
d) 提供一个有代表性的样本（最多1页）以说明数据集的属性/特征。

问题3：数据问题和预处理 (4分)
a) 识别数据集中的问题（例如，缺失、噪声、不一致数据）。（1 分）
b) 解释应用的预处理技术（例如，清理、转换、归一化）。（2 分）
c) 提供前后对比以展示改进效果。（1 分）

问题 4：数据挖掘技术（4 分）
a) 识别使用的数据挖掘技术（例如，Apriori、分类、聚类）。（1 分）
b) 说明选择该技术（或技术组合）解决该问题和数据集的原因。（1 分）
c) 简要描述该技术（术）是如何实现的。（2分）

问题5：模型验证（4分）
a) 描述该模型是如何被评估的。（1分）
b) 确定所使用的指标（例如，准确率、支持度、置信度、提升度）。（1分）
c) 解释为确保可靠性所使用的验证技术或实验设计。（2分）

Question 6: Conclusion and Future Improvements (3 Marks)
a) Summarize key findings and insights from the analysis. (1 Marks)
b) Discuss the significance of patterns, trends, and relationships identified. (1 Marks)
c) Highlight limitations and suggest improvements or future research directions. (1 Marks)

Question 7: Python Implementation (2 Marks)
a) Provide well-structured and documented Python code for preprocessing and analysis. (1 Marks)
b) Ensure reproducibility and clarity of implementation. (1 Marks)

Question 8: References (1 Mark)
a) List all references, tools, libraries, and academic sources used. (0.5 Marks)
b) Follow an appropriate citation style (e.g., APA). (0.5 Marks)

Question 9: Project Presentation and Viva (14 Marks)
a) Clarity and organization of presentation (4 Marks)
b) Understanding of project and methodology (4 Marks)
c) Explanation of results and ability to answer questions (4 Marks)
d) Team participation and communication will directly affect individual/group marks. (2 Marks)
Ideas for Projects / Reference

You can refer to this site for some project ideas: -

https://archive.ics.uci.edu/ml/datasets.html But you can get your data from any other suitable sites. Data sets: https://www.kaggle.com/

** Few more data sets are also used:
1. https://www.kaggle.com/code/onesinustamba/deep-learning-cnn-dcnn-for-brain-tumor-detection
2. https://www.kaggle.com/datasets/pkdarabi/medical-image-dataset-brain-tumor-detection/code
3. https://www.kaggle.com/code/saife245/melanoma-detail-analysis-eda-ip-augmentation-model
4. https://www.kaggle.com/code/sasakitetsuya/smoker-lung-cancer-stage-detection-model
5. https://www.kaggle.com/code/itsuki9180/mvan-covid-mrna-vaccine-analysis-notebook-268
6. https://www.kaggle.com/code/allunia/pulmonary-dicom-preprocessing

(Hint: Convert images data sets in CSV file using library of CSV)

- ***Students can use MONAI datasets and any publically available datasets also.***

Part-II Submission Requirements

1. Report Formatting:

- Font Type: Times New Roman
- Font Size: 12
- Line Spacing: 1.5
- Alignment: Justified

2. Report Structure:

- Page Limit: 5 – 25 pages (strictly adhere to this range).

Q问题 6：总结论断未来展望发现和见解。(1分)　(3 分
b) 讨论改进

b) 讨论已识别的模式、趋势和关系的意义。(1 分)
c) 突出局限性，并提出改进措施或未来研究方向。(1 分)

问题 7：Python 实现 (2 分)
a) 提供结构良好且文档齐全的 Python 代码，用于预处理和分析。(1 分)
b) 确保实现的可重复性和清晰性。(1 分)

问题 8：参考文献 (1 分)
a) 列出所有使用的参考文献、工具、库和学术来源。(0.5 分)
b) 遵循适当的引用格式（例如，APA 格式）。(0.5 分)

问题 9：项目展示和答辩 (14 分)
a) 演示的清晰度和组织性（4分）
b) 对项目和方法的理解（4分）
c) 结果的解释和回答问题的能力（4分）
d) 团队参与和沟通将直接影响个人/小组得分。（2分）
项目创意/参考

您可以参考此网站获取一些项目创意：-

https://archive.ics.uci.edu/ml/datasets.html 但你可以从任何其他合适的网站获取你的数据集：https://www.kaggle.com/

** 还使用了更多数据集：1. https://www.kaggle.com/code/onesinustamba/deep-learning-cnn-dcnn-for-brain-tumor-detection 2. https://www.kaggle.com/datasets/pkdarabi/medical-image-dataset-brain-tumor-detection/code 3. https://www.kaggle.com/code/saife245/melanoma-detail-analysis-eda-ip-augmentation-model 4. https://www.kaggle.com/code/sasakitetsuya/smoker-lung-cancer-stage-detection-model 5 268 6 9180 . https://www.kaggle.com/code/itsuki/mvan-covid-mrna-vaccine-analysis-notebook- . https://www.kaggle.com/code/allunia/pulmonary-dicom-preprocessing

(提示：使用CSV库将图像数据集转换为CSV文件)

- 学生可以使用***MONAI***数据集以及任何公开可用的数据集。

第二部分提交要求

1. 报告格式：

- 字体类型：Times New Roman
- 字体大小：12
- 行间距：1.5
- 对齐方式：两端对齐

2. 报告结构：

- 页数限制：5-25页（严格遵循此范围）。

- Cover Page: Must include Name, ID, Date, Signature, and Score.
- Marking Rubrics: Attach the grading rubric as the second page.
- Source Code: Include the full implementation of your sentiment analysis approach.
- Answer Script Report: Provide a structured report summarizing your analysis and findings.

Notes:

- Emphasize practical application.
- Brief explanations for each step.
- Use in-text citations and provide a list of references at the end (APA format).
- The Turnitin similarity score should be below 15% overall and less than 1% from a single source, excluding program source codes.

The following items are to be handed in:

- Submission of softcopy version to MyTIMeS
- Soft copy version of the *project report in PDF* and *DOC* version must be uploaded into My TiMES before your presentation schedule. Failed to do so, 30% will be deducted from your project

Question / Criteria	Marks	Excellent (Full Marks)	Good	Satisfactory	Needs Improvement / Poor
Introduction & Purpose of Study	4	Clear, concise context; well-defined problem; strong justification for data mining use and expected outcomes.	Good explanation with minor gaps in clarity or justification.	Basic description; problem or purpose not clearly articulated.	Unclear or missing context/problem; no justification.
Description of Data	4	Data source clearly identified; structure and attributes well explained; relevant sample provided.	Source and structure explained with minor missing details.	General dataset description; limited attribute explanation or sample.	Poor or missing dataset description.
Data Problems & Preprocessing	4	All major data issues identified; thorough preprocessing with clear before/after comparison.	Most issues identified; preprocessing applied with some explanation.	Limited discussion of issues; basic preprocessing shown.	No clear preprocessing or issues identified.
Data Mining Techniques	4	Appropriate technique clearly explained and strongly justified; implementation well described.	Technique explained with reasonable justification; minor gaps.	Basic explanation with weak justification.	Incorrect or unclear technique selection.
Model Validation	4	Clear validation approach; appropriate metrics (e.g., accuracy, support,	Validation explained with some metrics; minor clarity issues.	Basic validation with limited metric explanation.	No or incorrect validation approach.

- 封面页：必须包含姓名、ID、日期、签名和分数。
- 评分标准：将评分标准作为第二页附上。
- 源代码：包含您情感分析方法的完整实现。
- 答案脚本报告：提供一份结构化报告，总结您的分析和发现。

备注：

- 强调实际应用。
- 每一步的简要说明。
- 使用文中引用，并在文末提供参考文献列表（APA格式）。
- Turnitin相似度得分应总体低于15%，且来自单一来源的部分应少于1%，不包括程序源代码。

单个来源，不包括程序源代码。

以下物品需提交：

- 将软拷贝版本提交至MyTIMeS。
- 项目报告的软拷贝版本（PDF和DOC格式）必须上传到My TiMES系统中，在您的演示时间表之前。若未能完成，将扣除您项目30%的分数

问题 / 标准	分数	优秀（满分）	Good	合格	需要改进 / 差
研究介绍与目的	4	清晰简洁的背景；明确的问题；数据挖掘的充分理由和预期结果。	解释良好，但清晰度或理由存在轻微不足。	基本描述；问题或目的未明确阐述。不清晰或	缺失上下文/问题；no论证。
数据描述	4	数据来源明确标识；提供了相关的样本。量缺失细节。	结构和属性解释清晰；来源和结构解释存在少	一般数据集描述；属性解释或样本有限。数据集描述不佳或缺失。	
数据问题与预处理	4	所有主要数据问题已识别；进行了彻底的预处理，并进行了清晰的对比（前后）。大多数问题已识别；应用了预处理，并附有部分解释。		对问题讨论有限；展示了基本的预处理。	未识别出明确的预处理或问题。
数据挖掘技术	4	技术选择合理，解释清晰且充分论证；实施过程描述详细。	技术解释合理，论证基本充分；存在少量不足。	基本解释，但缺乏充分依据。	不正确或模糊不清技术选择。
模型验证	4	清晰的验证方法；合适的指标过一些指标解释；存	（例如，准确率、支持度、	基础验证与有限的指标说明。	无验证或验证方法不正确。

		confidence, lift) well explained.			
Conclusion & Future Improvements	3	Strong conclusions; meaningful insights; clear limitations and realistic future improvements.	Good conclusions with some insights and improvements.	General conclusions; limited depth or unclear improvements.	Weak or missing conclusions and future direction.
Python Implementation	2	Well-structured, clean, and fully documented code; reproducible results.	Code is functional and mostly clear; minor documentation issues.	Code provided but lacks clarity or structure.	Missing, incomplete, or non-functional code.
References	1	Complete and correctly formatted references (APA or appropriate style).	Minor formatting issues in references.	Incomplete references or inconsistent style.	Missing or incorrect references.

		置信度，提升率) 解释得很清楚。			
结论与未来改进	3	结论有力；有意义的见解；清晰的局限性以及未来实际改进方案。	结论良好，包含一些见解和改进。	总体结论；深度有限或改进方案不明确。薄弱或	缺失结论 and 未来方向。
Python 实现	2	结构良好，简洁，并完全文档化；可重复的结果。代码功能正常，大部分清晰；文档存在轻微问题。		提供了代码，但缺乏清晰度或结构。	缺失、不完整， or 非功能性代码。
参考文献	1	完整且格式正确的参考文献（APA或适当格式）。	轻微格式问题在参考文献中。	参考文献不完整或格式不一致。	缺失 或不正确的参考文献。

Assignment 3 Marking Rubrics

Criteria	Weightage	Beginning (0–49%)	Developing (50–64%)	Mastering (65–79%)	Excellent (80–100%)
Work collaboratively to design, execute, and complete a predictive modelling project, demonstrating accountability, autonomy, and ethical responsibility	10.00%	Demonstrates limited clarity, organization, and comprehensiveness. Objectives are unclear or poorly defined. Literature review is insufficient or irrelevant. Data description lacks detail. Methodology is weak or unjustified. Project plan is unrealistic or missing. Expected outcomes and impact are unclear. Challenges and limitations are not adequately addressed. Team roles and responsibilities are poorly defined.	Demonstrates moderate clarity and organization. Objectives are partially defined. Literature review is somewhat relevant. Data description is basic. Methodology is somewhat justified. Project plan is present but lacks detail. Expected outcomes and impact are somewhat described. Some challenges are identified. Team roles are partially defined.	Demonstrates good clarity, organization, and comprehensiveness. Objectives are well-defined and justified. Literature review is relevant. Data description is clear. Methodology is justified and feasible. Project plan is realistic. Expected outcomes and impact are clearly described. Challenges and limitations are addressed. Team roles and responsibilities are defined.	Demonstrates an exceptional level of clarity, organization, and comprehensiveness. Objectives are well-defined and strongly justified. Literature review is thorough and highly relevant. Data description is detailed and clear. Methodology is well-justified and feasible. Project plan and timeline are highly realistic and detailed. Expected outcomes and impact are clearly articulated. Challenges and limitations are critically addressed. Team roles and responsibilities are clearly and effectively defined.
Project Implementation, Data Processing, and Results Presentation	30.00%	Demonstrates limited execution and poor organization. Data preprocessing is inadequate or incorrect. Analysis is weak or incomplete. Results are unclear or poorly presented. Report lacks clarity and coherence.	Demonstrates basic execution with some organization. Data preprocessing is partially correct. Analysis is present but lacks depth. Results are somewhat clear. Report shows some structure but lacks consistency.	Demonstrates good execution and organization. Data preprocessing is appropriate. Analysis is meaningful. Results are clearly presented. Report is structured and understandable.	Demonstrates excellent execution and organization. Data preprocessing and model implementation are performed effectively. Results are clearly presented in a well-structured and coherent report. Insights are meaningful, with strong interpretation and clear community or business implications.

END OF ASSIGNMENT-3

作业3评分标准

标准	权重	开始 (0–49%)	发展中 (50–64%)	精通 (65–79%)	优秀（80–100%）
协作设计、执行并完成一个预测建模项目，展现责任感、自主性和道德责任	10.00%	展现有限的清晰度、组织性和全面性 . 目标表述不清或定义不明确。文献综述不足或不相关。数据描述缺乏细节。方法学薄弱或不合理。项目计划不切实际或缺失。预期成果和影响 are 不明确。挑战和局限性未得到充分解决。团队角色 and 职责 are poorly 未定义。	展现出适度的清晰度和组织性。目标部分定义。文献综述有一定相关性。数据描述基础。方法学有一定合理性。项目计划存在，但缺乏细节。预期成果和影响部分描述。部分挑战已识别。团队角色部分定义。	展示出良好的清晰度、组织性和全面性。目标明确且合理。文献综述相关。数据描述清晰。方法论合理可行。项目计划现实。预期成果和影响描述清晰。已应对挑战和局限性。团队角色和职责已定义。	展示了一个卓越的清晰度、组织性和全面性。目标明确且论证充分。文献综述深入且高度相关。数据描述详细清晰。方法论论证充分且可行。项目计划和时间表高度现实且详细。预期成果和影响阐述清晰。挑战和局限性得到批判性分析。团队角色和职责明确并且有效地定义了。
项目实施、数据处理和结果展示	30.00%	演示有限执行 and poor 组织。Data 预处理不足或不正确。分析薄弱或不完整。结果不清晰或质量差。已呈现。报告 缺乏清晰度 and 连贯性。	展示了基本的执行和一些组织。数据预处理部分正确。分析存在但缺乏深度。结果较为清晰。报告显示了一些结构但缺乏一致性。	展示了良好的执行和组织。数据预处理恰当。分析有意义。结果清晰呈现。报告结构合理且易于理解。	展示了 优秀 and 执行 组织 数据预处理和模型实现都高效完成。结果以结构清晰、逻辑连贯的报告呈现。见解具有意义，具有强大的解读能力，并具有明确的社区或商业影响。

作业-3结束